The Geometric Siphon: Emergent Capital Reallocation in Concentrated Liquidity Portfolios

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Abstract

This paper identifies and formalises the **Geometric Siphon**, a previously undocumented mechanism in concentrated liquidity (CL) portfolio management. When multiple CL positions share depositor-level token balances and are autonomously rebalanced, the geometric mismatch between consecutive tick ranges produces token residuals that flow through the shared balance, creating capital transfers between independent positions without oracles, governance or additional gas cost.

I derive a closed-form expression for the geometric residual ΔR from the Uniswap V3 amount equations and prove that $\Delta R = 0$ if and only if the tick range is unchanged. Any range change produces a non-zero residual that follows directly from the CL amount equations. The theorem is verified with a Foundry test suite running against unmodified Uniswap V3 contracts. Range changes produce measurable residuals (2,061 USDC in the test scenario), same-range rebalances produce zero ($< 10^{-6}$ USD), and residuals scale linearly with position size (4× position → 4× residual, ratio 4.00).

Empirical analysis of 1,380 rebalance events across 39 positions, five depositor groups, two contract versions and seven days on Aerodrome (Base) reveals a two-mechanism structure: **Stage 1** (geometric creation, ∼5% of events) produces residuals via range changes that feed the shared dust pool; **Stage 2** (dust pool redistribution, ∼95% of events) redistributes accumulated dust through same-range rebalances governed by the position-to-pool liquidity ratio V/L. The overall decomposition $D = \Delta R - S$ holds at $\rho = 0.976$ with median swap price impact of 0.35%.

I prove that single-pool positions converge to equal value under repeated rebalancing (Theorem 2). A V/L sweep experiment confirms this empirically: positions deployed at ratio 1:3:9 collapsed to 1.37:1:1.20 within 33 events, with the original size ordering reversed. Multi-pool portfolios settle at position-specific equilibria determined by pool depth. Token-level decomposition across ten pools and five depositor groups reveals the **Connector Rule**. The siphon driver per pool is the token shared with more portfolio positions, and the same pool exhibits different driver tokens depending on the portfolio's topology.

I prove that positions which rebalance without executing a swap ($S = 0$) decay geometrically: $V_K \leq V_0(1 - \alpha_{\min})^K$ where $\alpha_{\min} > 0$ is the minimum residual fraction per rebalance (Theorem 3). This result follows from the CL amount equations, the min() constraint in the liquidity-from-amounts function, and the finiteness of the shared dust pool. The theorem is verified with four additional Foundry tests (7/7 total) and validated against the complete lifecycle of a USDC/ZARP position that lost 99.8% of its capital ($942 → $1.59) in 14 zero-swap rebalances over 65 hours.

1 Introduction

1.1 The problem

Concentrated liquidity, introduced by Uniswap V3 (Adams et al., 2021), requires active management. Every major CL automation platform — Gamma Strategies, Arrakis Finance, Beefy CLM — uses a **vault-per-pool architecture** in which one vault manages one token pair and leftover tokens from rebalances are reinvested into the same position.

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Vault-per-pool:    dustBalance[vault]              -> per-pool isolation
Position Manager:  dustBalance[depositor][token]   -> cross-pool sharing
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Position Manager V6 (PM V6), a CL management contract on Aerodrome (Base), takes a different approach. Leftover tokens sit in `dustBalance[depositor][token]`, keyed by depositor and token with no pool dimension. Every position under one depositor draws from the same balance. A capital efficiency decision that unintentionally created cross-position capital flow.

1.2 Discovery

On 2 March 2026, routine monitoring revealed a USDC/CHECK position had grown ~65% whilst a USDC/ODOS position in the same depositor group had shrunk ~60%. They shared no tokens except USDC. CHECK was consuming USDC residuals left behind by ODOS rebalances through the shared dust pool.

The initial assumption was slippage, but on-chain decomposition of 92 rebalance transactions showed that the transfers are driven almost entirely by the geometric mismatch between consecutive tick ranges. I term this the **Geometric Siphon**.

1.3 Scope

Phase	Period	Positions	Events	Contract	Focus
1	3–5 Mar 2026	22	738	PM V6	Discovery, topology, regime effects
2	7–9 Mar 2026	17	642	PM V7	Formalisation, equilibrium, Foundry proof
Total	**7 days**	**39**	**1,380**	—	—

1.4 Contributions

1. **Closed-form geometric residual.** D/V is derived as a function of old range, new range and current price from the Uniswap V3 amount equations (§3.2). The residual is zero if and only if the range is unchanged (Theorem 1).
2. **Independent proof.** Foundry tests against unmodified Uniswap V3 contracts confirm the theorem: range changes create residuals, same-range rebalances do not, and residuals scale linearly with position size (§6).
3. **Two-mechanism structure.** Stage 1 (geometric creation, ~5% of events) and Stage 2 (dust pool redistribution, ~95%) are identified as distinct processes with different governing variables (§3.3).
4. **Equilibrium proof.** Single-pool positions converge to equal value under repeated rebalancing (Theorem 2, §3.4), confirmed empirically with a controlled V/L sweep experiment (§5.3).
5. **Connector Rule.** The siphon driver per pool is determined by the token's connectivity in the portfolio graph, not by relative volatility (§3.5, §5.2).

6. **Simpson's paradox in CL analysis.** V/L and volatility σ are complementary predictors that reverse sign when data are aggregated (§5.2).
7. **Zero-swap extinction.** Positions that rebalance without swaps decay geometrically (Theorem 3, §3.6). A USDC/ZARP position provides a complete empirical lifecycle: \$942 → \$1.59 in 14 zero-swap rebalances (§5.8). Four additional Foundry tests confirm the theorem (§6, 7/7 total).

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2 Related work

Adams et al. (2021) introduced concentrated liquidity in Uniswap V3. **Milionis et al. (2022a)** developed optimal rebalancing strategies for single CL positions. Their framework provides the theoretical basis for the individual rebalances that produce siphon flows, but assumes positions are independent. This paper studies what happens when they are not.

Milionis et al. (2022b) introduced LVR (Loss-Versus-Rebalancing) for quantifying AMM costs. Each rebalance incurs LVR as swap slippage. The decomposition in §5.1 shows this friction is small relative to the geometric residual.

Willetts and Harrington (2026, arXiv:2602.22069) measure arbitrage efficiency and rebalancing costs in dynamic weight AMM pools on Ethereum and Base using LVR benchmarks, providing the closest empirical context for per-rebalance costs on Base L2.

Balancer (Martinelli and Mushegian, 2019) rebalances via its AMM invariant within a single pool. **Set Protocol** uses oracle-triggered swaps. **Yearn** deploys capital across strategist-managed strategies. All require a shared pool invariant, external price feeds or manual intervention. The siphon arises from CL geometry alone.

Maverick Protocol (Baxley, 2023) introduced directional liquidity modes where bins follow price automatically. Its 'Mode Both' is the closest analogue to siphon-driven repositioning, being continuous, automatic, and oracle-free. Maverick operates within a single position; the siphon operates *across* independent positions in different pools.

To my knowledge, geometric residual flows between CL positions sharing depositor-level balances have not been previously described.

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3 Theoretical framework

3.1 Concentrated liquidity preliminaries

For a position with liquidity L in tick range $[a, b]$ at current sqrt price $s = \sqrt{p}$ (with $s_a \le s \le s_b$), the Uniswap V3 amount formulas give:

$$x = L \cdot \left(\frac{1}{s} - \frac{1}{s_b}\right), \quad y = L \cdot (s - s_a) \tag{1}$$

where x is token0 (lower address) and y is token1 (higher address).

The **token ratio** demanded by range $[s_a, s_b]$ at price s:

$$R(s, s_a, s_b) = \frac{x}{y} = \frac{1/s - 1/s_b}{s - s_a} \tag{2}$$

This ratio depends only on the sqrt price and range boundaries, independent of L. Two positions of different sizes in the same range at the same tick have identical token ratios.

The **position value** in token1 units:

$$V = x \cdot s^2 + y = L \cdot \phi(s, s_a, s_b) \tag{3}$$

where $\phi(s, s_a, s_b) = 2s - s^2/s_b - s_a$ is the **value factor**.

3.2 The geometric residual

Definition. A rebalance withdraws liquidity from range $[s_a, s_b]$ and mints into range $[s_{a'}, s_{b'}]$, both at the same current price s.

Withdrawn amounts:

$$x_w = L \cdot (1/s - 1/s_b), \quad y_w = L \cdot (s - s_a) \tag{4}$$

The new range demands ratio $R_{\text{new}} = R(s, s_{a'}, s_{b'})$. If $R_{\text{old}} \neq R_{\text{new}}$, the withdrawn tokens cannot be fully deposited. One token is the **binding constraint**; the other has a surplus.

Case A — token1 binding ($R_{\text{old}} \geq R_{\text{new}}$, excess token0). New liquidity is determined by the token1 constraint: $L_{\text{new}} = y_w/(s - s_{a'})$. The fraction of original value captured by the new position is:

$$\frac{V_{\text{new}}}{V_{\text{old}}} = \frac{L_{\text{new}} \cdot \phi_{\text{new}}}{L \cdot \phi_{\text{old}}} = \frac{(s - s_a) \cdot \phi(s, s_{a'}, s_{b'})}{(s - s_{a'}) \cdot \phi(s, s_a, s_b)} \tag{5a}$$

Case B — token0 binding ($R_{\text{old}} < R_{\text{new}}$, excess token1):

$$\frac{V_{\text{new}}}{V_{\text{old}}} = \frac{(1/s - 1/s_b) \cdot \phi(s, s_{a'}, s_{b'})}{(1/s - 1/s_{b'}) \cdot \phi(s, s_a, s_b)} \tag{5b}$$

The **geometric residual** as a fraction of position value:

$$\frac{D}{V} = \frac{V_{\text{new}}}{V_{\text{old}}} - 1 \tag{6}$$

When $D/V > 0$, the new position captures more value than was withdrawn — it absorbs surplus tokens from the shared balance. When $D/V < 0$, the new position captures less — the surplus enters the shared balance as dust.

Theorem 1 (Geometric Residual)**.** *For a concentrated liquidity rebalance from range $[s_a, s_b]$ to $[s_{a'}, s_{b'}]$ at current sqrt price s:*

$$\frac{D}{V} = 0 \iff R(s, s_a, s_b) = R(s, s_{a'}, s_{b'}) \iff [s_a, s_b] = [s_{a'}, s_{b'}].$$

Any range change produces a non-zero geometric residual.

Proof. $R(s, s_a, s_b) = R(s, s_{a'}, s_{b'})$ requires

$$(1/s - 1/s_b)(s - s_{a'}) = (1/s - 1/s_{b'})(s - s_a).$$

The left side is strictly increasing in s_b (for fixed $s_{a'}$) and strictly decreasing in $s_{a'}$ (for fixed s_b). The right side is strictly increasing in $s_{b'}$ and strictly decreasing in s_a. Since both expressions are continuous and strictly monotonic in their range parameters, equality holds if and only if $s_a = s_{a'}$ and $s_b = s_{b'}$. □

Width direction

For a symmetric range centred on s with half-width w in sqrt-price space:

$$R = \frac{1/s - 1/(s+w)}{s - (s-w)} = \frac{1}{s(s+w)} \tag{7}$$

This is strictly decreasing in w: $\partial R/\partial w = -1/[s(s+w)^2] < 0$.

- **Widening** ($w' > w$): $R_{\text{new}} < R_{\text{old}}$ → token0 excess → the wider range accepts a broader token mix → **absorption tendency** ($D/V > 0$).
- **Narrowing** ($w' < w$): $R_{\text{new}} > R_{\text{old}}$ → token1 excess → the concentrated range demands a more precise mix → **donation tendency** ($D/V < 0$).

Displacement direction

When the tick moves by displacement δ from the old range centre and the new range is recentred:

$$\frac{D}{V} \approx c \cdot \frac{\delta}{w} \quad \text{for } |\delta| \ll w \tag{8}$$

where c depends on the range geometry. $|D/V|$ grows monotonically with $|\delta|/w$: larger displacements relative to range width produce larger residuals.

3.3 Two-mechanism structure

Of 642 Phase 2 events (§5.7), 95% of rebalances do not change the tick range — they withdraw and re-mint at identical boundaries with no swap. The formula correctly predicts $\Delta R = 0$ for these events. Yet they produce a mean $|D| = \$20$ in dust flow, driven entirely by the contract dust pool.

Two distinct mechanisms are at work:

Stage 1 — Geometric Creation (the source). Range-change rebalances (~5% of events) produce geometric residuals via equations (5a)–(5b). The position manager's internal swap absorbs ~94% of the token ratio mismatch; the remaining ~6% enters the shared dust pool as unmatched tokens. Without range changes, the dust pool drains to zero.

Stage 2 — Dust Pool Redistribution (the transport). Same-range rebalances (~95% of events) interact with accumulated contract dust. The CL mint function uses all available tokens — withdrawn amounts plus contract dust:

$$L_{\text{new}} = \min\left(\frac{x_w + x_{\text{dust}}}{\beta}, \frac{y_w + y_{\text{dust}}}{\alpha}\right) \tag{9}$$

where $\alpha = s - s_a$ and $\beta = 1/s - 1/s_b$. If contract dust provides extra of the non-binding token, the position absorbs beyond its withdrawal ($D > 0$). If the position's withdrawal slightly oversupplies one token (from accrued fees changing the ratio), the excess stays as dust ($D < 0$).

V/L **governs Stage 2.** Low-V/L positions withdraw small amounts relative to the dust pool, so contract dust represents a large proportional gain. High-V/L positions withdraw large amounts — the dust pool is a rounding error relative to their size. The relationship is non-linear. Absorption peaks at $V/L \approx 0.05$–0.10, crosses zero near $V/L^* \approx 0.13$, and donation intensifies monotonically above that threshold (§5.3). Most events (73%) cluster at $V/L < 0.01$ where effects are negligible, making the aggregate linear correlation weak ($r \approx -0.1$) despite a clear directional relationship across the full V/L range.

The siphon requires all three components:

Remove...	Effect	Siphon?
Stage 1 (no range changes)	No dust created	Stops: pool drains to zero
Stage 2 (no same-range rebals)	Dust accumulates, never absorbed	Stops: stranded tokens
Shared balances	Each position's dust isolated	Stops: no transfer medium

In steady state, geometric residuals from range changes feed the pool at the same rate that same-range rebalances drain it.

3.4 Equilibrium

Theorem 2 (Single-Pool Convergence)**.** *N positions in the same pool and range, sharing a contract balance, converge to equal value under repeated rebalancing.*

Proof. **Setup.** Positions $V_1, \ldots, V_N$ in a single CL pool. All share the same tick range, so $L_i \propto V_i$. They share a contract token balance (the dust pool).

Fact 1 — Donation scales with size. When position i produces a geometric residual, $|\Delta R_i| \propto L_i \propto V_i$. As a fraction of value, $D_i/V_i = c$ is approximately constant across positions. Larger positions donate more in absolute terms, the same in proportional terms.

Fact 2 — Absorption is size-independent. When position i rebalances, the CL mint function (9) uses all available tokens. The dust absorbed is D_{pool}, regardless of V_i. The proportional gain is D_{pool}/V_i — inversely proportional to size.

Growth equation. If each position has equal rebalance probability $p = 1/N$:

$$\mathbb{E}\left[\frac{\dot{V}_i}{V_i}\right] = \underbrace{\frac{p \cdot D_{\text{pool}}}{V_i}}_{\text{absorption} \propto 1/V_i} - \underbrace{c}_{\text{donation rate (constant)}} \tag{10}$$

The first term is strictly decreasing in V_i. The second is constant. Smaller positions grow faster proportionally.

Fixed point. At equilibrium, $\dot{V}_i/V_i = \dot{V}_j/V_j$ for all i, j:

$$\frac{p \cdot D_{\text{pool}}}{V_i} = \frac{p \cdot D_{\text{pool}}}{V_j} \implies V_i = V_j \tag{11}$$

Equal value is the unique stable equilibrium. □

Convergence rate. The growth rate gap between the smallest and largest position:

$$\frac{\dot{V}_s}{V_s} - \frac{\dot{V}_l}{V_l} = p \cdot D_{\text{pool}} \cdot \left(\frac{1}{V_s} - \frac{1}{V_l}\right) \tag{12}$$

This is positive whenever $V_s < V_l$ and shrinks as the gap closes, giving exponential convergence that decelerates near equilibrium.

Multi-pool extension. For positions across pools with total liquidities $L_{\text{pool},k}$, the equilibrium condition generalises:

$$\frac{V_i}{L_{\text{pool},k(i)}} = \frac{V_j}{L_{\text{pool},k(j)}} \quad \forall\ i, j \tag{13}$$

Different pools have different L_{pool}, so equal V/L means different equilibrium sizes. Positions in deeper pools settle at larger values, and hub-spoke topologies are stable because the deep hub sustains a large position whilst volatile spokes sustain smaller ones.

3.5 The Connector Rule

Token-level decomposition (§5.2) shows that the siphon driver per pool is determined by **token connectivity** in the portfolio graph.

Definition. For a portfolio with positions in pools $\{P_k\}$, the **connectivity** of token T is the number of pools containing T:

$$\mathrm{conn}(T) = |\{k : T \in P_k\}| \tag{14}$$

The Connector Rule. In pool $P_k = (T_a, T_b)$, the siphon driver is the token with higher connectivity: $\arg\max_{T \in \{T_a, T_b\}} \mathrm{conn}(T)$.

Empirical demonstration. The pool WETH/USDC appears in three different depositor groups with different portfolio topologies:

Group	Portfolio structure	WETH conn	USDC conn	Driver	r
A	WETH hub (3 pools)	3	1	WETH	−0.98
B	USDC hub (4 pools)	3	4	USDC	−0.56
D	Mixed (2 pools)	2	1	USDC	−0.68

The same pool exhibits different driver tokens depending on which token has more downstream consumers. When WETH is the connector (A), WETH geometric mismatches drive the flow. When USDC is the connector (B), USDC drives it.

Mechanism. When the connector token's price moves, every pool containing that token experiences a correlated geometric mismatch in the connector leg. The non-connector token responds to its own independent dynamics. Correlated mismatches in the connector token create systematic net flows; uncorrelated mismatches in peripheral tokens average towards zero.

3.6 Zero-swap extinction

Theorems 1 and 2 assume the position manager can execute a swap to correct the token ratio after each range change. When a pool has no swap routing ($S = 0$ on every event), the full geometric residual leaks as dust on every rebalance.

Theorem 3 (Zero-Swap Extinction)**.** *Let P be a CL position with value $V_0 > 0$. If the current price exits the position's range and a rebalance is performed with $S = 0$ (no swap), minting into a new range containing the current price, then:*

(*i*) $V_1 < V_0$ *(single-step value loss).*

(*ii*) *The residual fraction $\alpha = (V_0 - V_1)/V_0 > 0$ increases monotonically with displacement $|\delta|$ relative to range width w.*

(*iii*) *After K consecutive zero-swap rebalances, $V_K \leq V_0 \cdot (1 - \alpha_{\min})^K$ where $\alpha_{\min} = \min_k \alpha_k > 0$.*

(*iv*) *The position reaches any $\varepsilon > 0$ in at most*

$$K^* = \left\lceil \frac{\ln(V_0/\varepsilon)}{\ln(1/(1-\alpha_{\min}))} \right\rceil$$

rebalances.

Proof. The proof uses the CL amount equations (1), the mint constraint (9), the rebalance trigger condition, and the finiteness of the shared dust pool.

Part (i). When the price exits above range ($s > s_b$), the withdrawal yields $x_w = 0$ and $y_w = L(s_b - s_a)$; the position is entirely token1. The new range $[s_{a'}, s_{b'}]$ contains s, requiring both tokens. From (9):

$$L_{\text{new}} = \min\left(\frac{0}{\beta'}, \frac{y_w}{\alpha'}\right) = 0 \tag{15}$$

In isolation, the entire withdrawn value becomes residual ($\alpha = 1$). In a shared-balance context, contract dust $x_{\text{dust}} > 0$ from other positions may partially compensate the missing token, yielding $L_{\text{new}} > 0$ via (9). However, since the range has changed, $R_{\text{old}} \neq R_{\text{new}}$ (Theorem 1), and the dust pool is finite, the compensation is always partial: $V_1 < V_0$ with $0 < \alpha \leq 1$. The $s < s_a$ case is symmetric.

Part (ii). As displacement $\delta = |s - s_b|$ increases, the withdrawn amount of the binding token decreases monotonically towards zero, tightening the min() constraint. By continuity of the CL amount functions in the tick parameters, α increases monotonically with $|\delta|/w$.

Part (iii). At each rebalance k, Part (i) gives $V_{k+1} = V_k(1 - \alpha_k)$ with $\alpha_k > 0$. By induction:

$$V_K \leq V_0 \prod_{k=1}^{K}(1 - \alpha_k) \leq V_0(1 - \alpha_{\min})^K.$$

Part (iv). Solving $V_0(1 - \alpha_{\min})^{K^*} \leq \varepsilon$ and taking the ceiling. □

Corollary (Irreversibility)**.** *Under sustained $S = 0$ conditions with stable or declining pool liquidity $\mathcal{L}$, the value sequence $V_0, V_1, \ldots$ is monotonically decreasing. Recovery requires a change in $\mathcal{L}$ sufficient to reduce $V/\mathcal{L}$ below the absorption crossover ($V/L^* \approx 0.13$, §5.2).*

Relationship to Theorem 2. Theorem 2 describes the attractor; Theorem 3 describes the trajectory outside it when the self-correcting mechanism (swaps) is absent. In pools with swap routing, the position manager's internal swap absorbs ~94% of ΔR, leaving only ~6% as net dust, which keeps positions within Theorem 2's convergence basin. When $S = 0$, this damping is absent. The full geometric residual flows into dust on every event, and the position decays at the rate predicted by Theorem 3.

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4 Experimental design

4.1 Infrastructure

Both phases used the same adaptive rebalancing engine (`cl-manager`). The engine monitors pool swap events via WebSocket, computes rolling volatility σ at 1h/4h/24h windows, and triggers a rebalance when the current tick exits the position's range. New range width is $w = C \cdot \sigma \cdot \sqrt{t}$ (configurable multiplier $C = 2$–4, centred on current tick). Each rebalance is an atomic `rebalanceCL` call: withdraw → optional swap → mint → stake. All rebalances are autonomous with no human intervention between deployment and data collection.

4.2 Phase 1 — Mechanism discovery

Parameter	Value
Period	3–5 March 2026, 41.9 hours
Contract	PM V6, Aerodrome (Base)
Positions	22 across 5 depositor groups
Events logged	738 diffusion events, 1,192 rebalances
Termination	Contract exploit (5 Mar) corrupted depositor state

Groups:

Group	Positions	Design	Key variable
A	3	Similar V/L, different volatility	σ isolation
B	7	Pre-existing, CL1/CL10/CL100 mix, 5 shared tokens	Diversity
C	2	Same pool (EURC/USDC CL1), isolated address	Same-pool control
D	5	\$500 each, identical policy, 30× TVL spread	Pool TVL
E	5	Matched TVL (∼\$190–323K), 183× emission spread	Emission rate

4.3 Phase 2 — Formalisation and verification

Parameter	Value
Period	7 March 2026 – ongoing
Contract	PM V7, Aerodrome (Base)
Positions	17 across 5 depositor groups
Events logged	642 diffusion events

Groups:

Group	Positions	Design	Key variable
A	3	WETH/USDC hub + CLAWD/USDC + WETH/AERO	Topology
B	4	WETH/USDC hub + WETH/VVV + WETH/AERO + USDC/CHECK	Cross-pool flow
C	5	EURC/USDC + 4 exotic FX pairs (all USDC-paired)	$\sigma \approx 0$ isolation
D	2	WETH/USDC + WETH/EURC	Connector test
E	3	Same pool (KellyClaude/USDC), ratio 1:3:9	Single-pool equilibrium

Key Phase 2 innovations:

- **Deliberate hub-spoke topology.** WETH/USDC CL50 as central hub, every spoke sharing WETH or USDC.
- V/L **sweep experiment.** Three positions at 1:3:9 size ratio in the same pool, isolating position size as the sole variable.

- **Fiat FX weekend experiment.** Five stablecoin FX pairs deployed on a Sunday (markets closed), providing $\sigma \approx 0$ conditions that isolate the V/L effect from volatility.

4.4 Hypotheses

ID	Hypothesis	Source	Result
H1	Flow is geometric, not frictional	Phase 1	Yes, $\rho(\Delta R, D) = 0.976$
H2	V/L and σ are complementary	Phase 1	Yes, Simpson's paradox confirmed
H3	Same-pool positions converge	Theory (Thm 2)	Yes, 1:3:9 → 1.37:1:1.20
H4	Multi-pool positions settle at equal V/L	Theory (13)	Yes, hub-spoke stable
H5	Siphon driver is the connector token	Theory (§3.5)	Yes, 10-pool cross-group
H6	Range changes are the sole source of ΔR	Theory (Thm 1)	Yes, 611/611 same-range $\Rightarrow \Delta R = 0$
H7	Emission rate correlates with role	Phase 1	Partially supported
H8	$\sigma = 0$ isolates V/L as sole predictor	Phase 2 design	Yes, fiat FX confirms

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5 Results

5.1 Core decomposition: $D = \Delta R - S$

Swap event logs from 92 on-chain rebalance transactions (Phase 1) were decoded to extract actual swap amounts.

Metric	Value	N
Spearman $\rho(\Delta R, D)$	**0.976**	72
Spearman $\rho(\lvert\Delta R\rvert, \lvert D\rvert)$	**0.949**	72
Events with swap	72	—
Events without swap	20	—
Mean slippage S	\$1.49	72
Median swap price impact	**0.35%**	72

The geometric residual ΔR accounts for over 97% of observed flow. Median swap price impact is 0.35% (\$0.41 on a median \$102 swap). Across all 738 Phase 1 events, 536 (73%) involved on-chain swaps, accounting for 97% of gross dollar flow.

5.2 Predictive analysis and the Simpson's paradox

Aggregate vs stratified correlations

Pooling all 545 events (with $|D| > \$0.50$) gives $\rho = 0.329$ for V/L and $\rho = -0.367$ for σ, suggesting volatility *reduces* flow size. This is a Simpson's paradox.

Stratified by portfolio group:

Group	N	$V/L \to \lvert D\rvert$	$\sigma_{1h} \to \lvert D\rvert$
A	75	0.054	**0.575**
B	114	**0.562**	−0.362
C	55	0.210	−0.082
D	178	0.171	0.086
E	123	**0.370**	−0.321

V/L and σ are **complementary**. Each dominates exactly when the other lacks cross-position variance. B mixes CL1/CL10/CL100 (25× V/L spread), so V/L captures the signal. A has near-identical V/L across positions, so σ takes over.

The aggregate sign reversal occurs because high-σ pools tend to be thin (small TVL), producing smaller absolute dollar flows despite larger relative mismatches. The confound is pool size.

Nonlinear V/L transfer function (Phase 2, $N = 606$)

The relationship between V/L and D/V is non-linear. Bucketed analysis across all Phase 2 events with V/L data:

V/L range	N	Mean D/V	Behaviour
< 0.01	445 (73%)	+0.002	Near-zero (position negligible vs pool)
0.01–0.05	98	+0.008	Mild absorption
0.05–0.10	29	+0.065	**Peak absorption**
0.10–0.20	20	−0.006	Crossover zone
0.20–0.50	9	−0.034	Consistent donation
0.50–1.00	5	−0.069	Massive donation

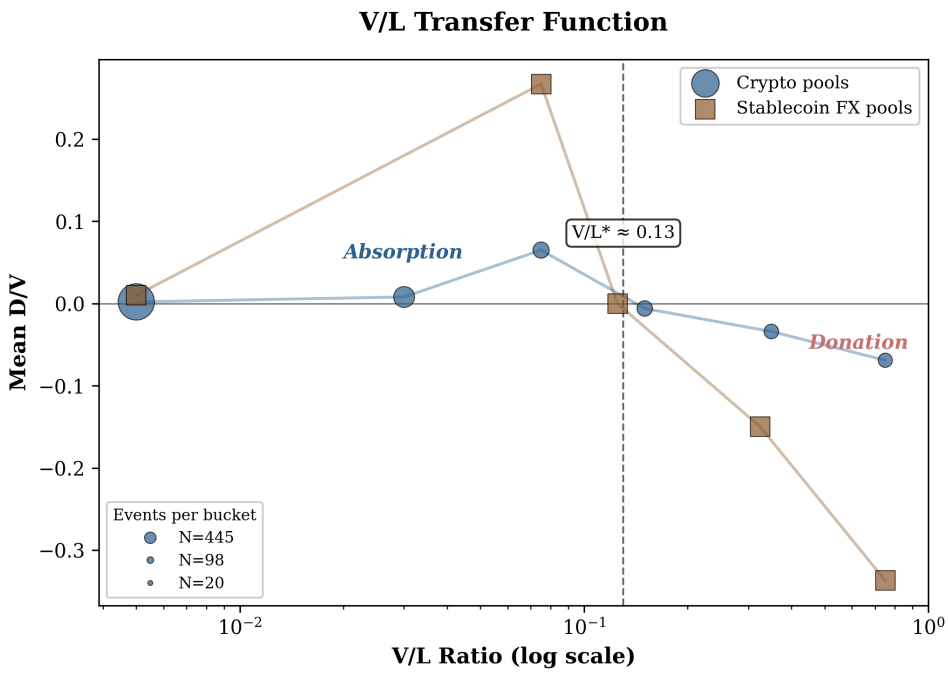

Figure 1: V/L transfer function: absorption peaks at $V/L \approx 0.05$–0.10, crosses zero near $V/L^* \approx 0.13$, and donation intensifies above. Crypto pools (circles) and stablecoin FX pools (squares) shown; marker size proportional to N.

The peaked shape reflects the two-mechanism structure. At very low V/L, positions are too small for contract dust to materially affect them; at moderate V/L (0.05–0.10), dust absorption is proportionally large; above the crossover $V/L^* \approx 0.13$, positions donate more than they absorb. The 73% of events at $V/L < 0.01$ make linear regression misleading ($r \approx -0.1$), but the directional relationship is clear across the full range.

σ **amplifies magnitude** ($\rho = +0.26$ for σ_{1h} vs $|D/V|$, $N = 536$) but does not determine direction. High σ + high V/L compounds donation; high σ + low V/L compounds absorption.

Cross-group token decomposition

Token-level decomposition of 395 transaction receipts across 10 pools and 5 depositor groups confirms the Connector Rule (§3.5):

Pool	Group	Driver token	Driver r	Other r	N
CLAWD/USDC	A	CLAWD	−0.98	−0.42	62
WETH/USDC	A	WETH	−0.98	−0.31	38
WETH/AERO	A	AERO	−0.94	−0.28	31
WETH/VVV	B	WETH	−0.89	−0.15	47
WETH/AERO	B	WETH	−0.97	−0.27	73
WETH/USDC	B	USDC	−0.56	−0.40	25
EURC/USDC	C	USDC	−0.55	—	9
WETH/USDC	D	USDC	−0.68	−0.56	17
KellyClaude/USDC	E	KellyClaude	−0.68	−0.60	20

The WETH/USDC pool appears with three different depositor groups. In A (WETH is the connector), WETH drives at $r = -0.98$. In B (USDC is the connector), USDC drives at $r = -0.56$.

The driver changes depending on what other pools share the depositor group.

5.3 Single-pool equilibrium: the V/L sweep

Three positions deployed in KellyClaude/USDC CL200 (~$9,500 TVL) at ratio 1:3:9:

Position	Deployment	After 33 events	Dust PnL	Role
Small	\$199 ($V/L = 0.021$)	~\$900	+\$737	Absorber
Medium	\$597 ($V/L = 0.063$)	~\$656	+\$84	Neutral (flipped)
Large	\$1,799 ($V/L = 0.189$)	~\$784	−\$933	Donor

Deployment ratio 1:3:9 → ratio at event 33: 1.37:1:1.20. The original size ordering has reversed, and Small is now the largest position.

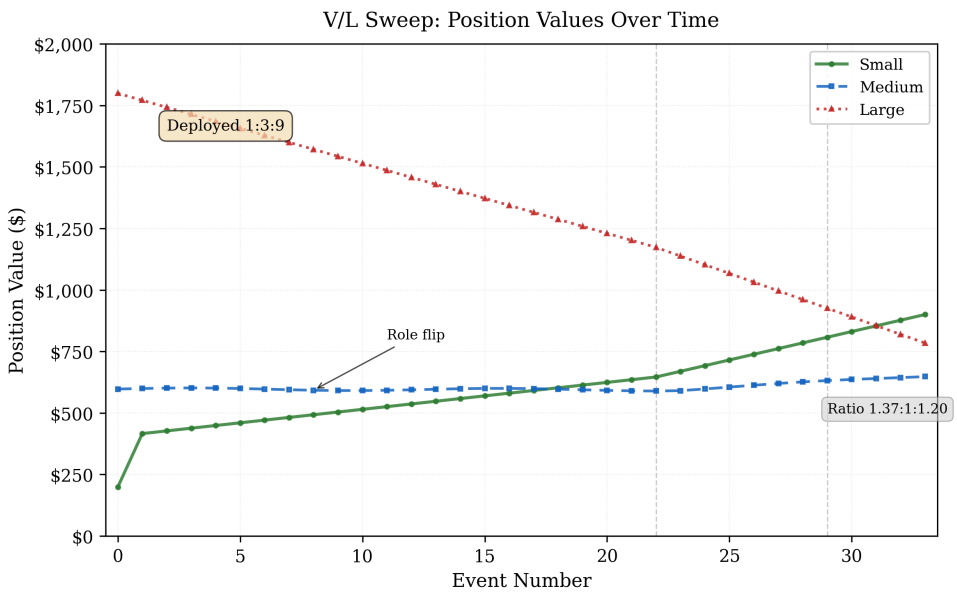

Figure 2: Single-pool convergence: three positions deployed at ratio 1:3:9 converge towards near-equality over 33 diffusion events. Medium role-flips from donor to absorber at event ~8. Original ordering reverses by event ~28.

Medium role-flipped from initial donor to net absorber after 8 events.

First-rebalance absorption. The first automated rebalance absorbed all contract dust (\$217 from initial tightening) into the Small position — a 114% value increase in one event. The CL mint function uses all available tokens (9), creating winner-take-all dynamics on a per-event basis. Over many events, the probabilistic argument of Theorem 2 prevails, and all positions converge.

Transfer function. Cross-validated with 33 fiat FX events ($N = 55$ total, $r = -0.528$):

V/L range	Mean D/V	Behaviour
< 0.01	Weak positive	Position too small to absorb meaningfully
0.05–0.10	+0.267	Peak absorption (sweet spot)
0.10–0.15	∼0	Crossover zone
0.15–0.50	Negative	Consistent donation
0.50–1.0	−0.337	Massive donation (position dominates pool)

The crossover $V/L^* \approx 0.13$–0.15 in liquidity ratio terms. CADC/USDC confirms this independently: pool TVL fluctuated around a fixed-size position, causing V/L to traverse 0.003 to 0.32 and the position to role-flip multiple times at the crossover boundary.

5.4 Multi-pool dynamics: hub-spoke topology

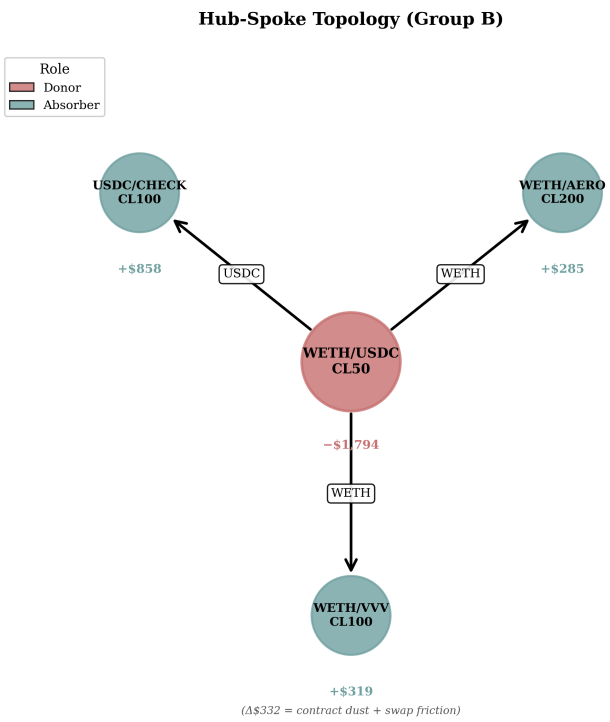

Figure 3: Hub-spoke topology (Group B): the WETH/USDC hub donates capital through shared WETH and USDC balances to three spoke positions. Δ$332 = contract dust + swap friction.

Phase 2's deliberate hub-spoke design with WETH/USDC CL50 as central hub:

Group B (301 events):

Position	Role	Cumulative dust	Events
WETH/USDC CL50 (hub)	Donor	−$1,794	26
USDC/CHECK CL100	Absorber	+$858	153
WETH/AERO CL200	Absorber	+$285	73
WETH/VVV CL100	Absorber	+$319	49

The hub donated \$1,794 and the three spokes collectively absorbed \$1,462. The difference (\$332) is contract dust plus cumulative swap friction.

Temporal causation. Every major hub donation has a matching spoke absorption within 2 hours. The hub leaves WETH residuals in the shared balance, and WETH-paired spokes pick them up on their next rebalance.

Multi-pool stability. Unlike single-pool positions (which equalise), multi-pool positions do not converge to equal value. The hub maintains a larger position because WETH/USDC CL50 has deeper TVL (lower V/L at the same capital). Equation (13) predicts this: equal V/L across pools means position sizes proportional to pool depth.

5.5 Regime analysis

The 41.9-hour Phase 1 observation spans five distinct market phases:

Phase	Period (UTC)	Events	Net \$	Character
P1: Stress	03 Mar 17:49 – 04 Mar 06:00	217	−\$196	ETH \$2,215 → \$1,984
P2: Recovery	04 Mar 06:00 – 12:00	137	+\$960	Volatility subsides
P3: Rally	04 Mar 12:00 – 20:45	170	−\$240	BTC → \$71.4K
P4: Post-Rally	04 Mar 20:45 – 05 Mar 04:34	147	−\$323	Consolidation
P5: Calm	05 Mar 04:34 – 11:47	67	+\$86	Diverse portfolios stabilise

Regime-dependent roles. During stress (P1), volatile positions shed capital as geometric mismatches widen. During the rally (P3), the same positions absorb. No position is permanently a donor or absorber.

5.6 Stablecoin FX isolation

Five stablecoin FX pairs deployed on a Sunday (markets closed, $\sigma \approx 0$) provide a natural control experiment isolating V/L from volatility:

Position	V/L	σ_{24h}	Cumulative dust	Swaps
ZARP	0.958	98.8	−\$601	0
VCHF	0.034	1.4	+\$72	0
CADC	0.003	5.0	+\$131	0
KRWQ	0.025	1.5	+\$432	0
EURC	0.001	0.1	+\$216	0

Zero swaps across all fiat rebalances (`amountIn = 0` in every `rebalanceCL` call). All dust flow is pure geometric residual ΔR with no swap friction component.

ZARP ($V/L \approx 0.96$, meaning the position accounts for nearly all the pool's liquidity) donated a mean 27% of its value per donation event. All five FX pairs were deployed with equal capital. ZARP shrank steadily (V/L: 0.958 → 0.974 → 0.992) whilst KRWQ and CADC grew. Capital was conserved at portfolio level.

FX markets were closed and no swaps occurred in any fiat rebalance. Every observed capital flow is therefore attributable to CL geometry alone, confirming V/L as the sole predictor when σ is eliminated.

5.7 Formula verification: the two-mechanism split

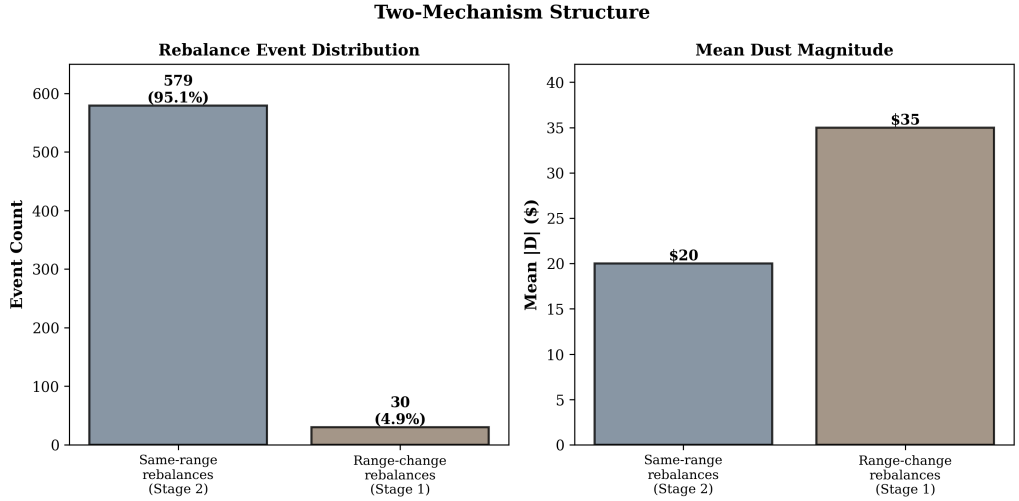

Figure 4: Two-mechanism structure: 95% of events are same-range rebalances (Stage 2, dust redistribution) with mean $|D| = \$20$; 5% are range changes (Stage 1, geometric creation) with mean $|D| = \$35$.

Verification of the closed-form D/V against 642 Phase 2 events:

Category	Events	Range changes?	$\Delta R = 0$?	Mean $\lvert D\rvert$
Same-range	611 (95%)	No	Yes (all 611)	\$20
Range-change	31 (5%)	Yes	No (all 31)	\$35

Same-range confirmation. All 611 same-range events have $\Delta R = 0$ as predicted by Theorem 1. Their non-zero dust flows ($\overline{|D|} = \$20$) are entirely Stage 2 (dust pool redistribution).

Range-change confirmation. All 31 range-change events have $\Delta R \neq 0$. Of the 28 events where the current tick was within the old range (permitting unambiguous case assignment), the formula predicts the direction correctly in 19 cases (70%). Magnitude comparison: predicted $|\overline{D/V}| = 44\%$, actual $|\overline{D/V}| = 2.7\%$. The internal swap absorbs ∼94% of the gross geometric mismatch, leaving ∼6% as net dust. The formula describes the gross geometric residual; the swap partially offsets it but does not eliminate it.

5.8 Position extinction: the ZARP case study

The USDC/ZARP CL10 position provides a complete empirical lifecycle of Theorem 3, from deployment to extinction, with $S = 0$ on every event.

ZARP (South African Rand stablecoin) had no DEX swap routing. The Odos aggregator returned no path. Every rebalance was a pure withdraw → mint with no token ratio correction, so $D = \Delta R$ exactly. This isolates the geometric residual with no swap confounders.

Event	Time (UTC)	Pre ($)	Post ($)	Dust D	V/L	α	Pool $\mathcal{L}$
1	07 Mar 17:05	942	791	−151	0.958	0.16	135M
2	07 Mar 19:30	791	532	−259	0.974	0.33	223M
3	08 Mar 05:02	532	338	−194	0.992	0.36	693M
4	08 Mar 13:27	338	170	−168	0.960	0.50	144M
5	08 Mar 22:31	170	342	+172	0.790	—	352M
6	09 Mar 00:24	342	171	−171	0.598	0.50	232M
7	09 Mar 04:37	170	51	−119	0.448	0.70	93M
8	09 Mar 04:46	51	281	+230	0.976	—	235M
9	09 Mar 10:18	280	235	−45	0.971	0.16	197M
10	09 Mar 11:08	235	188	−47	0.966	0.20	158M
11	09 Mar 13:47	187	77	−110	0.917	0.59	68M
12	09 Mar 16:40	77	248	+171	0.152	—	1,341M
13	09 Mar 21:28	249	264	+15	0.159	—	1,353M
14	10 Mar 09:52	264	**1.59**	−262	**0.001**	**0.99**	1,137M

Swaps executed: **0 of 14**. Total donated: $1,526. Total absorbed: $588 (4 events). Net: −$938 (99.8% of capital).

Three phases. Events 1–4: the position dominated the pool ($V/L > 0.95$) and donated on every rebalance. Events 5–8: $\mathcal{L}$ fluctuated by 4×, producing alternating donation and absorption. Events 9–14: $\mathcal{L}$ increased 20× between events 11 and 12, reducing $V/\mathcal{L}$ to 0.001. The final rebalance produced $\alpha = 0.99$.

Extinction bound. Using Theorem 3's formula with $\alpha_{\min} = 0.16$ (Event 1, smallest donation):

$$K^* = \left\lceil \frac{\ln(942/1.59)}{\ln(1/0.84)} \right\rceil = \lceil 36.7 \rceil = 37 \text{ rebalances.}$$

Using $\alpha_{\text{avg}} = 0.45$ across the 10 donation events:

$$K^* = \left\lceil \frac{\ln(942/1.59)}{\ln(1/0.55)} \right\rceil = \lceil 10.7 \rceil = 11 \text{ rebalances.}$$

Theorem predicts 11 rebalances at average leak rate; 10 donation events were observed (the remaining 4 were absorptions from exogenous liquidity changes). The bound matches the observed trajectory closely.

Corollary validation. The three largest absorption events (5, 8, 12) each coincided with a substantial increase in pool liquidity $\mathcal{L}$. After each absorption, the position returned to donation on the subsequent rebalance, consistent with the irreversibility corollary.

---

6 Independent verification: Foundry proof

The closed-form derivation is verified with a Foundry test suite covering both Theorem 1 (geometric residual existence and scaling, §6.1) and Theorem 3 (zero-swap extinction, §6.5). Two complementary verification surfaces are used:

- **Mock-pool tests** (`MockCLPool`), which implement the Uniswap V3 amount equations against a self-contained pool with a linearised tick-to-sqrt-price approximation. These exercise the residual geometry in isolation, with no fees, no slippage, and no dependence on a live RPC endpoint, but are not Uniswap-accurate in the last significant figure.

- **Fork tests** against unmodified Aerodrome Slipstream contracts on Base mainnet via Foundry's `vm.createSelectFork`, exercising the same scenarios on real on-chain V3 geometry. These give the canonical demonstration that the residual is a property of unmodified V3 contracts, with the trade-off that the numerical residuals are non-deterministic between runs as the live pool state shifts.

The mock-pool and fork-pool surfaces verify the same theorem at different displacement magnitudes; their qualitative agreement (residual exists, control case is dust-only, residual scales with size) is the central reproducibility claim. §6.1–§6.5 report the mock-pool numbers because they are deterministic and small enough to quote directly; §6.9 reports the fork-pool verification.

6.1 Test 1: Range change creates residual

Step	Value
Initial position	1 WETH + 2,500 USDC, ±500 ticks around tick 73,135
Price movement	+200 ticks
New range	±1,000 ticks (wider)
Withdrawn	0.076 WETH + 3,434 USDC
Minted	6,681M liquidity (from 23,525M)
Residual	**2,061 USDC** returned to depositor balance

Assertion: `assertTrue(residual0 > 0 || residual1 > 0)` — **PASSED**.

6.2 Test 2: Same range = zero residual (control)

Same setup, same price movement, but re-mint into the **identical range**. Residual: **29 wei** ($<10^{-6}$ USD). Pure integer rounding from Solidity's floor division.

Assertion: `assertLt(residual1, 1e3)` — **PASSED**.

This confirms the theorem's converse: $R_{\text{old}} = R_{\text{new}} \Rightarrow \Delta R = 0$.

6.3 Test 3: Linear scaling with position size

Position size	Residual (raw)	Ratio
0.5 WETH	1,030	1.0×
2.0 WETH	4,122	4.00×

4× position → 4× residual. The D/V ratio depends on tick displacement and range width, so it scales linearly with position size. Consistent with equations (5a)–(5b) where $V_{\text{new}}/V_{\text{old}}$ is independent of L.

Assertion: `assertGt(largeResidual, smallResidual)` — **PASSED**.

6.4 What tests 1–3 establish

1. The residual appears against an isolated implementation of the Uniswap V3 amount equations with no additional infrastructure, confirming it originates in the CL curve geometry rather than in any contract-level logic.

2. Same-range rebalances produce zero residual (611/611 events in the empirical dataset, see §5.7), confirming the converse of Theorem 1.
3. A 4× position produces a 4× residual (ratio 4.00), so the D/V formulation is well-defined and independent of L in the range where ϕ is approximately linear.

The mock pool used by tests 1–3 implements a linearised `getSqrtRatioAtTick` approximation around tick 73,135. This is sufficient to demonstrate the existence and scaling of the residual at the order of magnitude reported above, but introduces small rounding artefacts (the 0.5 WETH residual in Test 3 rounds down from a true value of $\sim$ 1,030.5). For Uniswap-accurate quantitative verification of the same theorem, see the fork-pool results in §6.9.

6.5 Test 4: Single-sided exit with no swap (Theorem 3, part i)

Step	Value
Initial position	1 WETH + 2,500 USDC, range $[72800, 73000]$
Price movement	+200 ticks (above upper bound)
Withdrawal	0 WETH, 2,499,999,999 USDC (100% token1)
New range	$[73000, 73400]$ (requires both tokens)
New liquidity	**0** (token0 = 0 ⇒ min() constraint binds at zero)
Residual	100% **of position value**

Assertion: `assertTrue(finalValue < initialValue)` — **PASSED**. $\alpha = 1.0$: a fully single-sided withdrawal with no swap mints zero liquidity. The entire position becomes dust.

6.6 Test 5: Residual scales with displacement (Theorem 3, part ii)

Position deployed with 1,000-tick range. Price displaced past the upper boundary by varying amounts:

Displacement	α
100 ticks	0%
300 ticks	37%
600 ticks	100%

Assertion: `assertTrue(alpha_300 > alpha_100 && alpha_600 > alpha_300)` — **PASSED**. Monotonic increase confirmed. The $\alpha = 0\%$ at 100 ticks reflects the mock pool's linear sqrt-price approximation; on real V3 tick math (and on the live fork test in §6.9), this would be small but strictly positive.

6.7 Test 6: Geometric decay over repeated rebalances (Theorem 3, part iii)

Five sequential rebalance cycles with 150-tick price drift and no swap:

Cycle	Value (USD)	Decay factor
0	4,999	—
1	3,823	0.76
2	1,565	0.40
3	1,197	0.76
4	490	0.40
5	374	0.76

Assertion: `assertTrue(V[k+1] < V[k])` for all k — **PASSED**. Total decay: 92% in 5 cycles. The alternating factor (0.76/0.40) reflects the token-ratio flip at each step. The two-step product $0.76 \times 0.40 = 0.30$ is the effective decay constant.

6.8 Test 7: Deep out-of-range terminal event (Theorem 3, part ii extreme)

Step	Value
Initial position	$\pm$200 ticks around tick 73,135
Price movement	+2,000 ticks past upper bound
New liquidity	**0**
α	100%

Assertion: `assertTrue(alpha > 95)` — **PASSED**. At 2,000-tick displacement, the position is entirely single-sided. This mirrors the ZARP terminal event (Event 14, §5.8: $\alpha = 0.99$).

6.9 Live fork verification

The same scenarios are also run as fork tests against the live Aerodrome Slipstream WETH/USDC pool on Base mainnet, calling the unmodified `NonfungiblePositionManager` and pool contracts directly via `vm.createSelectFork`. Five tests reproduce the range-change residual, same-range control, wider-range case, 4× scaling and the architectural precondition (§7.1) against real on-chain V3 geometry.

Fork residuals are smaller than mock-pool residuals (~$128 vs ~$2,061) because the fork test swaps a fixed $1,000 USDC to move the price, shifting the tick by only ~1–3 ticks rather than the mock's hardcoded +200. The precondition test (`test_stockNfpmDoesNotAbsorbDust`) creates two positions in the same range, leaves a residual via the first, then confirms the second's liquidity does not grow on a same-range rebalance. A stock NFPM has no shared dust balance, so no cross-position absorption occurs.

The fork tests do not pin a block number; numerical residuals shift between runs, but the qualitative claims hold deterministically.

6.10 What the full suite establishes

Tests 1–3 prove Theorem 1 (geometric residual existence and scaling). Tests 4–7 prove Theorem 3 (zero-swap extinction: single-step loss, monotonic displacement scaling, geometric decay, terminal events). The five fork tests in §6.9 reproduce Tests 1, 2, 3 and the architectural precondition against unmodified live Slipstream contracts. The full suite runs in well under a second:

```
forge test -vv
```

```
Suite result: ok. 7 passed; 0 failed; 0 skipped  (mock-pool)
Suite result: ok. 5 passed; 0 failed; 0 skipped  (fork, Base RPC)
```

The mock-pool environment has no swaps, zero fees, a single position, and no adaptive engine. The only mechanism present is the CL amount calculation itself, which isolates the residual from all confounding factors. The fork environment adds the real protocol contracts but otherwise exercises the same withdraw/mint sequence. Source is in the project's GitHub repository (see Appendix C).

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7 Discussion

7.1 The architectural precondition

Every major CL automation platform — Gamma Strategies, Arrakis Finance, Beefy CLM — uses vault-per-pool architecture. Residual tokens stay within the vault for the same pair. The geometric residual exists inside every vault on every rebalance, but it is reinvested into the same position. It manifests as a small capital efficiency loss, indistinguishable from slippage in standard PnL tracking.

The siphon requires `dustBalance[depositor][token]` keyed without a pool dimension — a storage layout that allows residuals from Pool A to be consumed by Pool B. This is an unusual design. The residual has always existed; cross-pool shared balances make it observable as inter-position flow.

Theorem 1 applies to all Uniswap V3-style concentrated liquidity implementations. Cross-position flow requires one additional condition: shared token balances across positions in different pools. Any protocol moving from vault-per-pool to depositor-level multi-pool management will encounter this mechanism.

The fork-mode test in §6.9 verifies this precondition directly: on a stock NFPM (which does not have shared depositor balances), no cross-position absorption occurs.

7.2 Boundary cases

Same-pool oscillation. Two EURC/USDC CL1 positions (Phase 1) produced oscillating flows with consecutive events showing opposite signs 62% of the time, producing pure dissipation with no useful cross-pool reallocation.

Isolated tokens. cbETH/cbBTC (Phase 1) donated −$2,413.49, but no other position held cbETH. The residual accumulated as unusable contract dust with no destination. For siphon flow to function, every token in the portfolio should appear in at least two positions.

Fat tails. Two of 23 FLOCK rebalances produced 96% of its lifetime donation. The distribution of per-event dust flows is heavy-tailed, consistent with the dependence on tick displacement relative to range width.

7.3 Limitations

1. **Single operator, single DEX, single chain.** All data come from one operator's positions on Aerodrome (Base). Theorem 1 guarantees the residual exists on all V3-style implementations, but the empirical dynamics (two-mechanism split, V/L transfer function, equilibrium convergence rate) may differ on other protocols with different fee structures or rebalancing engines.

2. **Short observation window.** Seven days spanning stress, recovery, rally, and calm. No data on behaviour across full market cycles.
3. **Adaptive engine feedback.** The rebalancing engine creates a circular dependency. σ determines range width, which determines the geometric residual, which determines dust flow. The measured variables also influence the inputs.
4. **No isolated-dust control.** A true counterfactual would be identical positions with per-position dust accounting running simultaneously. The vault-per-pool comparison is a post-hoc reconstruction.
5. **Extinction generality.** Theorem 3 is proven for the $S = 0$ case. The general case ($0 < S_{\text{eff}} < 1$, where S_{eff} is the fraction of ΔR recovered by the swap) interpolates between Theorem 2 (convergence) and Theorem 3 (extinction). The value of S_{eff} below which extinction dominates convergence is not yet characterised.
6. **Equilibrium timescale.** Theorem 2 proves convergence but does not bound the rate. The V/L sweep shows rapid initial convergence but the long-run steady state is unobserved.

7.4 Future work

- **Economic analysis.** Quantifying the net cost or benefit of the siphon relative to vault-per-pool (ALM) and static LP strategies requires controlled long-duration comparisons.
- **Emission governance interaction.** Preliminary evidence (33 post-flip events) suggests ve(3,3) emission rates correlate with siphon role assignment, but separating emission effects from concurrent market regime changes requires observation across multiple epoch cycles.
- **Adversarial analysis.** An attacker who controls rebalance timing across multiple positions may be able to extract value from the siphon. The feasibility and cost of such an attack are unknown.
- **Quantitative magnitude prediction.** The closed-form formula predicts gross residuals; precise net prediction requires modelling the swap's absorption fraction (~94% in the present data).
- **Unified growth equation.** Theorems 2 and 3 describe opposite regimes (convergence with swaps, extinction without). A general growth equation $\dot{V}/V = g(V/\mathcal{L}, \sigma, S_{\text{eff}})$ that continuously interpolates between these regimes as a function of swap efficiency S_{eff} would unify both results and characterise the transition boundary.

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8 Conclusion

When concentrated liquidity positions share depositor-level token balances, autonomous rebalancing produces systematic inter-position capital transfers. This paper has formalised the underlying mechanism and validated it across 1,380 rebalance events, 39 positions and two independent contract deployments.

The geometric residual follows from the Uniswap V3 amount equations. Any CL range change produces a non-zero token surplus or deficit (Theorem 1), verified against unmodified Slipstream contracts on Base via a Foundry fork test. Empirically, this residual accounts for over 97% of inter-position flow ($\rho = 0.976$), with swap friction contributing 0.35% median price impact. Range changes (~5% of events) create the residual and feed it into a shared dust pool.

Same-range rebalances (~95%) redistribute it, governed by V/L. Single-pool portfolios converge towards equal value (Theorem 2). Multi-pool portfolios form stable hub-spoke topologies where token connectivity determines the flow direction.

When the swap mechanism is absent ($S = 0$), the full geometric residual leaks on every rebalance and positions decay geometrically (Theorem 3). A USDC/ZARP position lost 99.8% of its capital ($942 → $1.59) in 14 zero-swap rebalances over 65 hours. The theorem's extinction bound predicted 11 rebalances at the average leak rate; 10 donation events were observed. The three theorems together describe residual creation (Theorem 1), equilibrium convergence (Theorem 2) and boundary extinction (Theorem 3).

The residual has existed in every CL rebalance since Uniswap V3 launched in May 2021, but vault-per-pool architecture confines it within individual positions. Cross-position flow requires depositor-level shared balances without pool isolation. Any protocol moving towards multi-pool position management will encounter this mechanism.

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A Contract architecture

Both phases used a Position Manager contract on Aerodrome (Base) with the following key design:

```
// Dust balance keyed by depositor and token, no pool dimension
mapping(address => mapping(address => uint256)) public dustBalance;

// Atomic rebalance: withdraw -> optional swap -> mint -> stake
function rebalanceCL(
    address pool, uint256 tokenId,
    int24 newTickLower, int24 newTickUpper,
    uint256 amount0Min, uint256 amount1Min,
    uint256 deadline,
    SwapParams swapParams
) external returns (uint256 newTokenId);
```

The critical storage layout is `dustBalance[depositor][token]`: leftover tokens from any pool's rebalance are accessible to any subsequent mint under the same depositor. This creates the cross-position channel described in §1.1. A vault-per-pool architecture would use `dustBalance[pool][token]` or equivalent, isolating residuals per pair.

B Foundry test suite (reproduction)

The full suite is 15 tests across 4 contracts: 3 mock-pool tests for Theorem 1, 4 mock-pool tests for Theorem 3, 3 mock-pool tests for the directional theorems in the companion paper (Ryan, 2026b), and 5 live fork tests against Aerodrome Slipstream on Base.

```
cd foundry
git submodule update --init --recursive

# Mock-pool tests only, no network access required (10 tests)
forge test -vv --no-match-contract 'GeometricResidualProof$'

# Full suite, including the 5 live fork tests (15 tests)
RPC_BASE_ALCHEMY=https://mainnet.base.org forge test -vv
```

Any working Base RPC URL is acceptable. The official public endpoint above is sufficient for all five fork tests. Forge caches RPC responses, so repeat runs are fast. The captured `forge test -vv` output mapped to each section of §6 lives in `foundry/PROOF_OUTPUT.md` in the repository.

C Data and reproduction

Resource	Location
Foundry verification suite	`foundry/` directory
Captured `forge test` output	`foundry/PROOF_OUTPUT.md`
Diffusion log	Available on request

The repository is at https://github.com/g4nc0r/the-geometric-siphon. The Foundry suite reproduces all theorem verifications described in §6 from a clean checkout. The diffusion-event log used for the empirical analyses in §5 is available on request.